


Lecture # 8

Project Integration Management

ENGINEERING PROJECT MANAGEMENT

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Learning Outcomes

- Describe an overall framework for project integration management
  - Process & practices,
  - Inputs,
  - Outputs,
  - Tools & techniques

Project Integration Management

- **1. Project Integration Management.** Includes the processes and activities to identify, define, combine, **unify**, and **coordinate** **the** **various** **processes** and project management **activities** within the Project Management Process Groups.

Table 1-4. Project Management Process Group and Knowledge Area Mapping

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	

Project Integration Management - Processes

- **4.1 Develop Project Charter—**The process of developing a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities.
- **4.2 Develop Project Management Plan—**The process of defining, preparing, and coordinating all plan components and consolidating them into an integrated project management plan.
- **4.3 Direct and Manage Project Work—**The process of leading and performing the work defined in the project management plan and implementing approved changes to achieve the project's objectives.
- **4.4 Manage Project Knowledge—**The process of using existing knowledge and creating new knowledge to achieve the project's objectives and contribute to organizational learning

Project Integration Management - Processes

- **4.5 Monitor and Control Project Work—**The process of tracking, reviewing, and reporting overall progress to meet the performance objectives defined in the project management plan.
- **4.6 Perform Integrated Change Control—**The process of reviewing all change requests; approving changes and managing changes to deliverables, organizational process assets, project documents, and the project management plan; and communicating the decisions.
- **4.7 Close Project or Phase—**The process of finalizing all activities for the project, phase, or contract.

Develop Project Charter

- The process of developing a document that formally authorizes the existence of a project and provides the project manager with the authority to apply organizational resources to project activities.

Develop Project Charter

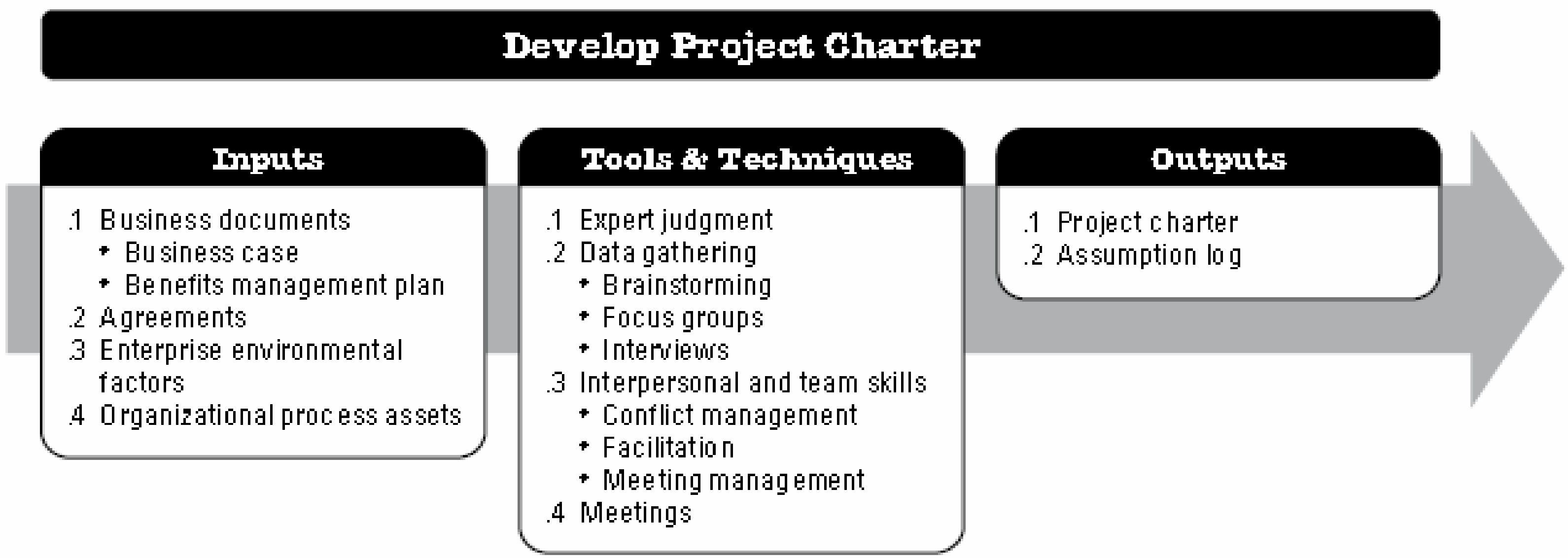

Figure 4-2. Develop Project Charter: Inputs, Tools & Techniques, and Outputs

Project Charter Steps

- Project purpose;
- Measurable project objectives and related success criteria;
- High-level requirements;
- High-level project description, boundaries, and key deliverables;
- Overall project risk;
- Summary milestone schedule;
- Preapproved financial resources;
- Key stakeholder list;
- Project approval requirements (i.e., what constitutes project success, who decides the project is successful, and
- who signs off on the project);
- Project exit criteria (i.e., what are the conditions to be met in order to close or to cancel the project or phase);
- Assigned project manager, responsibility, and authority level; and
- Name and authority of the sponsor or other person(s) authorizing the project charter.

Develop Project Charter - Template

PROJECT CHARTER			
Project Name			
Project Description			
Project Manager		Sponsor	
Authority Level of the Project Manager		Signature	
		Date Approved	
Business Case			
Estimated Budget			
Project Objectives			
Project Approval Requirements			
High Level Product Description / Key Deliverables			
Preassigned Resources			

Key Stakeholders	
Stakeholder Requirements	
High-Level Assumptions	
High-Level Constraints	
High-Level Risks	
Summary Milestones	
Project Exit Criteria	
Project Success Criteria	

Example

Project Title: DNA-Sequencing Instrument Completion Project
Date of Authorization: February 1
Project Start Date: February 1 **Projected Finish Date:** November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated $1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The DNA-sequencing instrument project has been underway for three years. It is a crucial project for our company. This is the first charter for the project, and the objective is to complete the first version of the software for the instrument in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

Example

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the DNA sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com

Sign-off: (Signatures of all the above stakeholders)

Ahmed Abrams *Nick Carson*
Susan Johnson *Renyong Chi*
Erik Haus *Bill Strom*
Maggie Elliot

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

"I want to be heavily involved in this project. It is crucial to our company's success, and I expect everyone to help make it succeed." —Ahmed Abrams

"The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me." —Renyong Chi

Project Management Plan

- The key benefit of this process is the production of a comprehensive document that defines the basis of all project work and how the work will be performed.

Develop Project Management Plan

Develop Project Management Plan

Inputs

.1 Project charter
.2 Outputs from other processes
.3 Enterprise environmental factors
.4 Organizational process assets

Tools & Techniques

.1 Expert judgment
.2 Data gathering
- Brainstorming
- Checklists
- Focus groups
- Interviews

.3 Interpersonal and team skills
- Conflict management
- Facilitation
- Meeting management

.4 Meetings

Outputs

.1 Project management plan

Table 4-1. Project Management Plan and Project Documents

Project Management Plan	Project Documents	
1. Scope management plan	1. Activity attributes	19. Quality control measurements
2. Requirements management plan	2. Activity list	20. Quality metrics
3. Schedule management plan	3. Assumption log	21. Quality report
4. Cost management plan	4. Basis of estimates	22. Requirements documentation
5. Quality management plan	5. Change log	23. Requirements traceability matrix
6. Resource management plan	6. Cost estimates	24. Resource breakdown structure
7. Communications management plan	7. Cost forecasts	25. Resource calendars
8. Risk management plan	8. Duration estimates	26. Resource requirements
9. Procurement management plan	9. Issue log	27. Risk register
10. Stakeholder engagement plan	10. Lessons learned register	28. Risk report
11. Change management plan	11. Milestone list	29. Schedule data
12. Configuration management plan	12. Physical resource assignments	30. Schedule forecasts
13. Scope baseline	13. Project calendars	31. Stakeholder register
14. Schedule baseline	14. Project communications	32. Team charter
15. Cost baseline	15. Project schedule	33. Test and evaluation documents
16. Performance measurement baseline	16. Project schedule network diagram	
17. Project life cycle description	17. Project scope statement	
18. Development approach	18. Project team assignments	

Simplified PM Plan

Project Management Plan

Project Life Cycle

Development Approach

Management Reviews

Performance Measurement Baseline

- Scope Baseline
- Cost Baseline
- Schedule Baseline

Change Management Plan + Configuration Management Plan

Scope Management Plan + Requirements Management Plan

Schedule Management Plan

Cost Management Plan

Quality Management Plan

Resource Management Plan

Communications Management Plan

Risk Management Plan

Procurement Management Plan

Stakeholder Management Plan

Change Management Plan

CHANGE MANAGEMENT PLAN

Project Name/ID:

Change Management Approach:

DEFINITIONS OF CHANGE

CHANGE	SCOPE:	
	SCHEDULE:	
	BUDGET:	
	PROJECT DOCUMENTS:	

CHANGE CONTROL BOARD

Name	Position	Responsibility	Authority

CHANGE CONTROL PROCESS

CHANGE REQUEST	Submital	
	Tracking	
	Review	
	Disposition	

Direct and Manage Project Work

- The key benefit of this process is that it provides overall management of the project work and deliverables, thus improving the probability of project success.

Direct and Manage Project Work

Direct and Manage Project Work

Inputs

.1 Project management plan
- Any component

.2 Project documents
- Change log
- Lessons learned register
- Milestone list
- Project communications
- Project schedule
- Requirements traceability matrix
- Risk register
- Risk report

.3 Approved change requests

.4 Enterprise environmental factors

.5 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Project management information system

.3 Meetings

Outputs

.1 Deliverables

.2 Work performance data

.3 Issue log

.4 Change requests

.5 Project management plan updates
- Any component

.6 Project documents updates
- Activity list
- Assumption log
- Lessons learned register
- Requirements documentation
- Risk register
- Stakeholder register

.7 Organizational process assets updates

Figure 4-6. Direct and Manage Project Work: Inputs, Tools & Techniques, and Outputs

Direct and Manage Project Work

ISSUE LOG								
Issue #	Issue Description	Impact Level	Report Date	Reported By	Responsible	Due Date	Status	Notes
001	The client is insisting for work that is not covered in the scope	High	2-Mar-18	John C.	William L.	2-Apr-18	Closed	
002	Key project team member is not working out	Medium	3-May-18	Marry P.	Brad H.	4-Jun-18	Open	Waiting for the approval of the executive management
003	User name allows HTML Injection	High	5-Jun-18	George W.	Tom F.	7-Jul-18	Closed	

Manage Project Knowledge

- The key benefits of this process are that prior organizational knowledge is leveraged to produce or improve the project outcomes, and knowledge created by the project is available to support organizational operations and future projects or phases.

Manage Project Knowledge

Manage Project Knowledge

Inputs

.1 Project management plan
- All components

.2 Project documents
- Lessons learned register
- Project team assignments
- Resource breakdown structure
- Source selection criteria
- Stakeholder register

.3 Deliverables

.4 Enterprise environmental factors

.5 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Knowledge management

.3 Information management

.4 Interpersonal and team skills
- Active listening
- Facilitation
- Leadership
- Networking
- Political awareness

Outputs

.1 Lessons learned register

.2 Project management plan updates
- Any component

.3 Organizational process assets updates

Figure 4-8. Manage Project Knowledge: Inputs, Tools & Techniques, and Outputs

Manage Project Knowledge

LESSONS LEARNED REGISTER				
TECHNICAL				
Date	Submitted by	Experience Type	Description of the problem/situation	Proposed Solution
MANAGEMENT				
Date	Submitted by	Experience Type	Description of the problem/situation	Proposed Solution
PROJECT MANAGEMENT				
Date	Submitted by	Experience Type	Description of the problem/situation	Proposed Solution

Manage and Control Project Work

- The key benefits of this process are that it allows stakeholders to understand the current state of the project, to recognize the actions taken to address any performance issues, and to have visibility into the future project status with cost and schedule forecasts.

Manage and Control Project Work

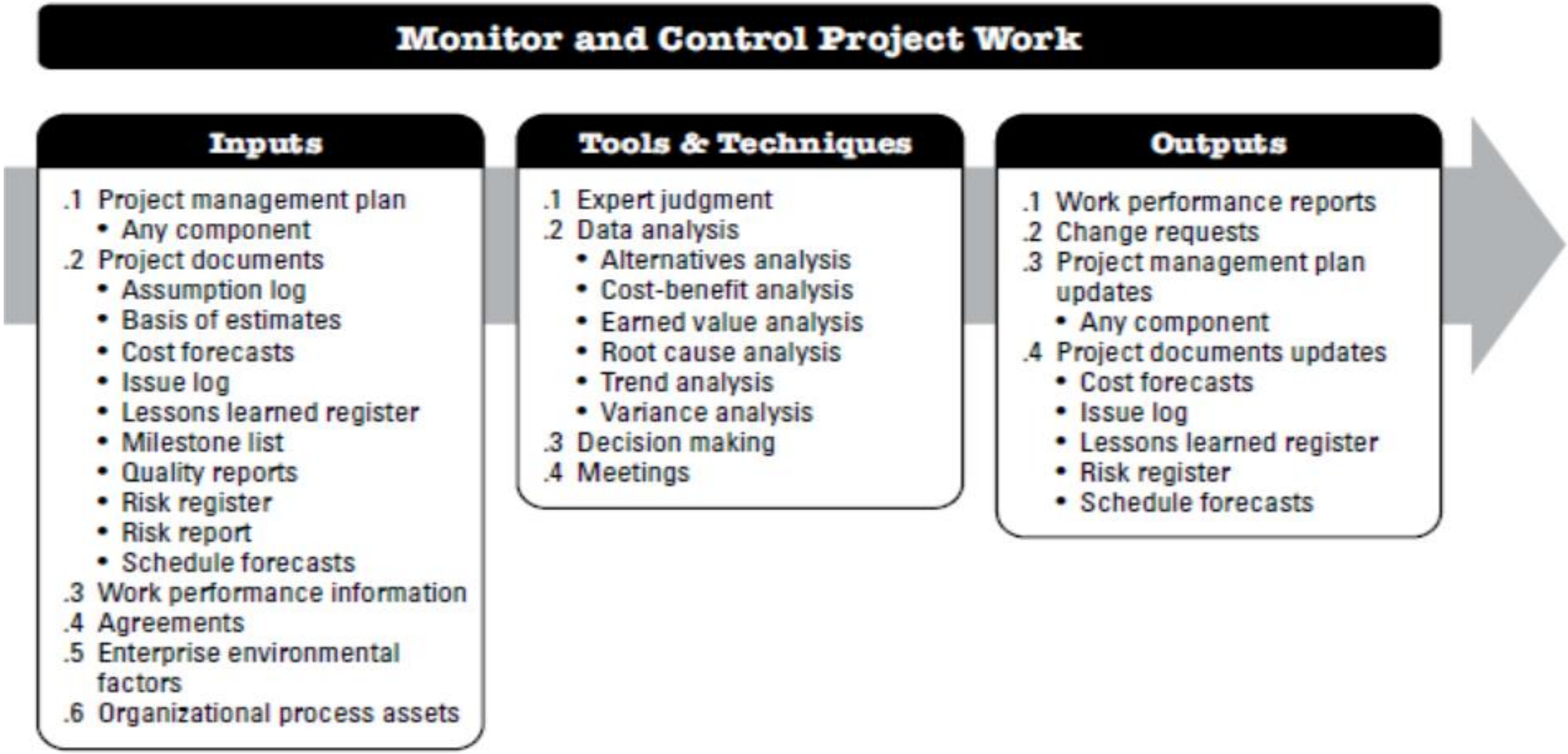

Figure 4-10. Monitor and Control Project Work: Inputs, Tools & Techniques, and Outputs

Manage and Control Project Work

What should you do if you find out a variation as a result of monitoring and controlling activities?

1) Search for the problem that causes the variation
2) Find out the root cause of the problem
3) Solve the root cause of the problem
4) Point out the works that may need corrective action
5) Investigate the impact of the change within the related knowledge area
6) Request corrective action (Change request)
7) Perform Integrated Change Control (We will see in the following lectures)

Manage and Control Project Work

(Performance) Status/Progress Report

Project Name: **Team Member Name:** **Date:** **Reporting Period:**
Work completed this reporting period:
Work to complete next reporting period:
What's going well and why:
What's not going well and why:
Suggestions/Issues:
Project changes

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Perform Integrated Change Control

- The key benefit of this process is that it allows for documented changes within the project to be considered in an integrated manner while addressing overall project risk, which often arises from changes made without consideration of the overall project objectives or plans.

Perform Integrated Change Control

Perform Integrated Change Control

Inputs

.1 Project management plan
- Change management plan
- Configuration management plan
- Scope baseline
- Schedule baseline
- Cost baseline

.2 Project documents
- Basis of estimates
- Requirements traceability matrix
- Risk report

.3 Work performance reports

.4 Change requests

.5 Enterprise environmental factors

.6 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Change control tools

.3 Data analysis
- Alternatives analysis
- Cost-benefit analysis

.4 Decision making
- Voting
- Autocratic decision making
- Multicriteria decision analysis

.5 Meetings

Outputs

.1 Approved change requests

.2 Project management plan updates
- Any component

.3 Project documents updates
- Change log

Figure 4-12. Perform Integrated Change Control: Inputs, Tools & Techniques, and Outputs

Perform Integrated Change Control

Change Request Form

Project Name:
Date Request Submitted:
Title of Change Request
Change Order Number:
Submitted by: (name and contact information)
Change Category: □**Scope** □**Schedule** □**Cost** □**Technology** □**Other**

Description of change requested:

Events that made this change necessary or desirable:

Justification for the change/why it is needed/desired to continue/complete the project:

Impact of the proposed change on:
Scope:
Schedule:
Cost:
Staffing:
Risk:
Other:

Suggested implementation if the change request is approved:

Required approvals:

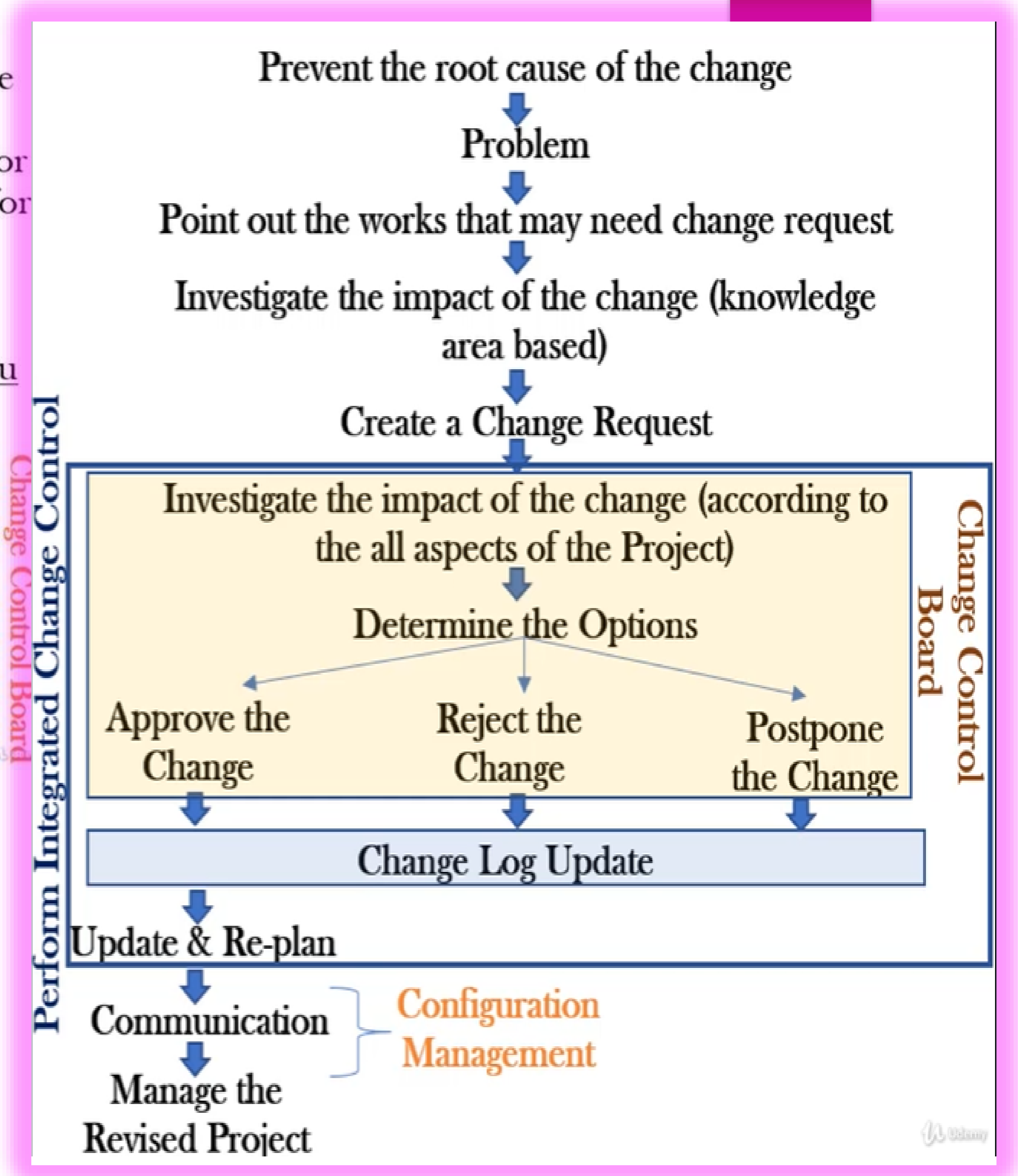
Name/Title	Date	Approve/Reject

Evaluation of the Change Request

Ex: The project you are involved is exceeding the budget. To cut off the costs, the project manager of the project decided to decrease the number of engineers working for the project. He made a Change Request for this reason. He made all the cost related calculations for this request and he comes up with this result: The project will save almost $1,000,000 if the number of engineers would be decreased from 10 to 5. As a member of the Change Control Board, how would you evaluate this situation?

You should investigate the impacts of this change on other aspects of the Project!

- Schedule
- Quality
- Scope
- Customer Satisfaction
- Risks
- Resource Motivation, etc.



Close Project or Phase

- The key benefits of this process are the project or phase information is archived, the planned work is completed, and organizational team resources are released to pursue new endeavors.

Close Project or Phase

Close Project or Phase

Inputs

.1 Project charter
.2 Project management plan
  • All components
.3 Project documents
  • Assumption log
  • Basis of estimates
  • Change log
  • Issue log
  • Lessons learned register
  • Milestone list
  • Project communications
  • Quality control measurements
  • Quality reports
  • Requirements documentation
  • Risk register
  • Risk report
.4 Accepted deliverables
.5 Business documents
  • Business case
  • Benefits management plan
.6 Agreements
.7 Procurement documentation
.8 Organizational process assets

Tools & Techniques

.1 Expert judgment
.2 Data analysis
  • Document analysis
  • Regression analysis
  • Trend analysis
  • Variance analysis
.3 Meetings

Outputs

.1 Project documents updates
  • Lessons learned register
.2 Final product, service, or result transition
.3 Final report
.4 Organizational process assets updates

Figure 4-14. Close Project or Phase: Inputs, Tools & Techniques, and Outputs